

Course Title: Introduction to Mechanical Engineering

Course Code: 22ESC144/244

Course In charge: Ashoka E

Module-3

Introduction to IC Engines: Components and Working Principles, 4-Stroke Petrol and Diesel Engines, Application of IC Engines.

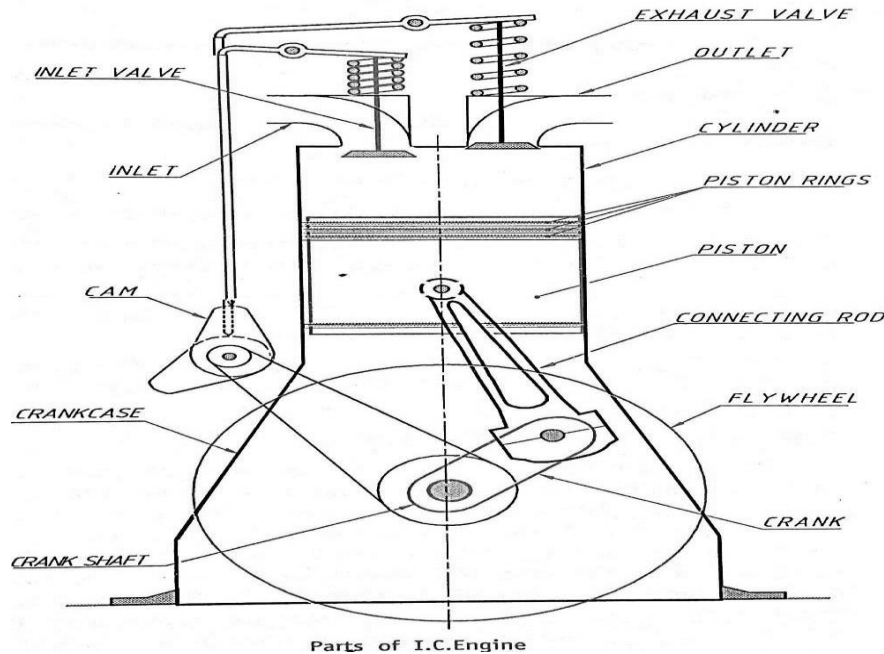
I.C. Engines

Working principle of IC engine

An Internal combustion engine called i.e., IC Engine, is a heat engine which converts heat energy released by the combustion of fuel taking place inside the engine cylinder into mechanical work. It has advantages such as high efficiency, light weight, compactness, easy starting, adaptability, comparatively lower initial cost has made its use as a prime mover.

Parts of I.C. Engines:

1. **Cylinder:** The heart of the engine is the cylinder in which the fuel is burnt and the power is developed. The inside diameter is called bore. To prevent the wearing of cylinder block, a sleeve will be fitted tightly in the cylinder. The piston reciprocates inside the cylinder.
2. **Piston:** The piston is a close fitting hollow cylindrical plunger moving to-and-fro in the cylinder. The power developed by the combustion of the fuel is transmitted by the piston to the crankshaft through the connecting rod.
3. **Piston rings:** The piston rings are the metallic rings inserted into the circumferential grooves provided at the top end of the piston. These rings maintain a gas-tight joint between the piston and the cylinder while the piston is reciprocating in the cylinder. They also help in conducting the heat from the piston to the cylinder.
4. **Connecting rod:** It is a link that connects the piston and the crankshaft by means of pin joints. It converts the rectilinear motion of the piston into rotary motion of the crankshaft.
5. **Crank and crankshaft:** The crank is a lever that is connected to the end of the connecting rod by a pin joint with its other end rigidly connected to a shaft called crankshaft. It rotates about the axis of the crankshaft and causes the connecting rod to oscillate.



6. Crank case: It is the lower part of the engine serving as an enclosure for the crankshaft and also sump for the lubricating oil.

7. Valves: The valves are the devices which controls the flow of the intake and the exhaust gas to and from the cylinder. They are also called poppet valves. These valves are operated by means of cams driven by crankshaft through a timing gear and chain.

8. Fly wheel: It is a heavy wheel mounted on the crankshaft of the engine to maintain uniform rotation of the crankshaft.

4-STROKE PETROL ENGINE: (S. I. Engine)

Petrol engines consists of cylinder, piston, connecting rod, crank, crankshaft, inlet valve, exhaust valve and spark plug. The spark plug fitted at the top of the cylinder initiates the ignition of the petrol, hence the name spark ignition engine.

1. SUCTION STROKE:

During this stroke the piston moves from TDC to BDC. The inlet valve is open and exhaust valve is closed. The crankshaft rotates by half a rotation. As the piston moves downwards, suction is created in the cylinder, as a result, fresh air-petrol mixture is drawn into the cylinder through the inlet valve. At the end of this stroke, the piston is in BDC, the cylinder is filled with air-petrol mixture and inlet valve closes. Horizontal line AB on the P-V diagram.

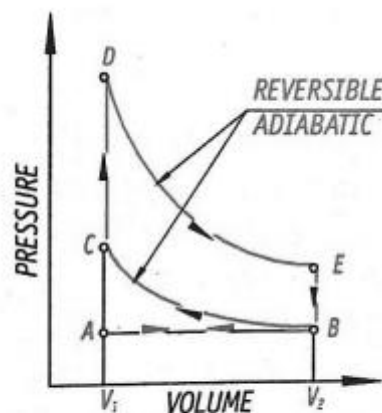
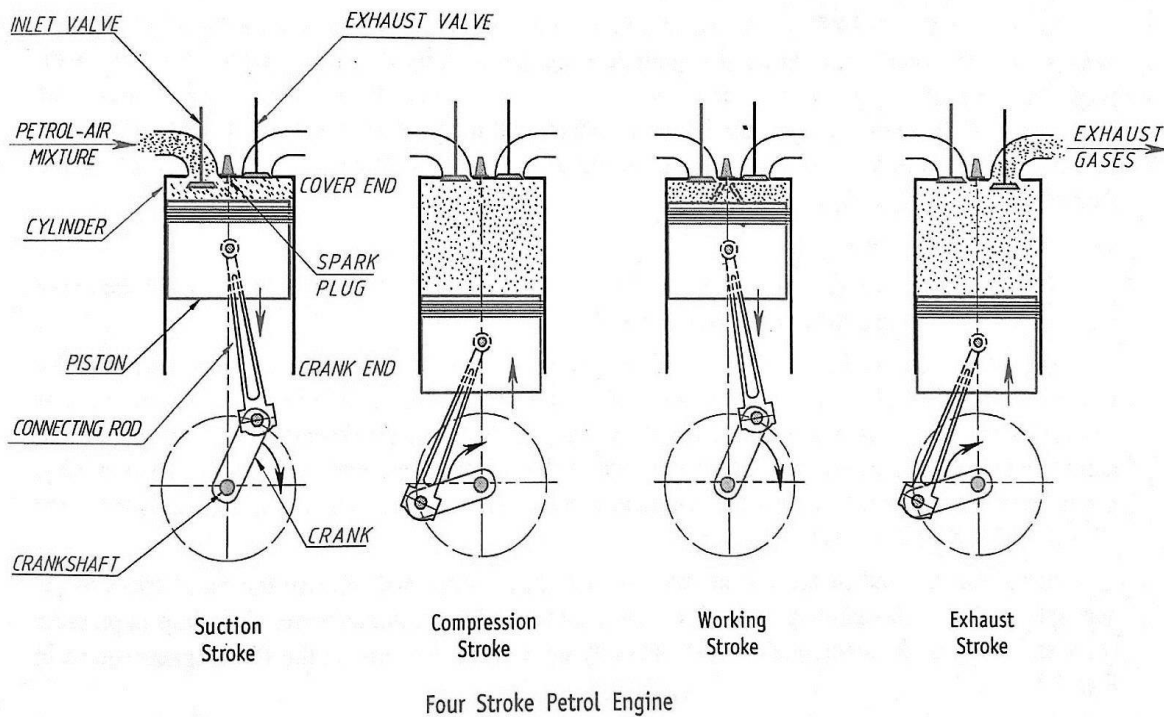
2. COMPRESSION STROKE:

During this stroke the piston moves from BDC to TDC. Both the inlet valve and exhaust valves are closed. The crankshaft rotates by half a rotation. As the piston moves upwards, the fuel mixture in the cylinder will be compressed. It is represented by the BC curve in the P-V diagram.

When the piston reaches TDC, the spark plug ignites the fuel mixture. Since the spark plug ignites the fuel (air-petrol), this type of engine is also called as spark ignition or S.I Engine. The combustion of fuel takes place increasing the pressure at constant volume, represented by the line CD in the P-V diagram.

3. WORKING OR POWER STROKE:

During this stroke the piston moves from TDC to BDC. Both the inlet valve and exhaust valves are closed. The crankshaft rotates by half a rotation. The high pressure of the burnt gases forces the piston downwards performing power stroke. The linear motion of the piston is converted to rotary motion of the crankshaft by connecting rod and crank. It is represented by curve on DE on PV diagram. At the end of the stroke, the piston is in BDC, the exhaust valve opens which release the burnt gases to the atmosphere. This will bring pressure in the cylinder to atmospheric at constant volume, represented by the line EB in the P-V diagram.



Theoretical Otto Cycle

4. EXHAUST STROKE:

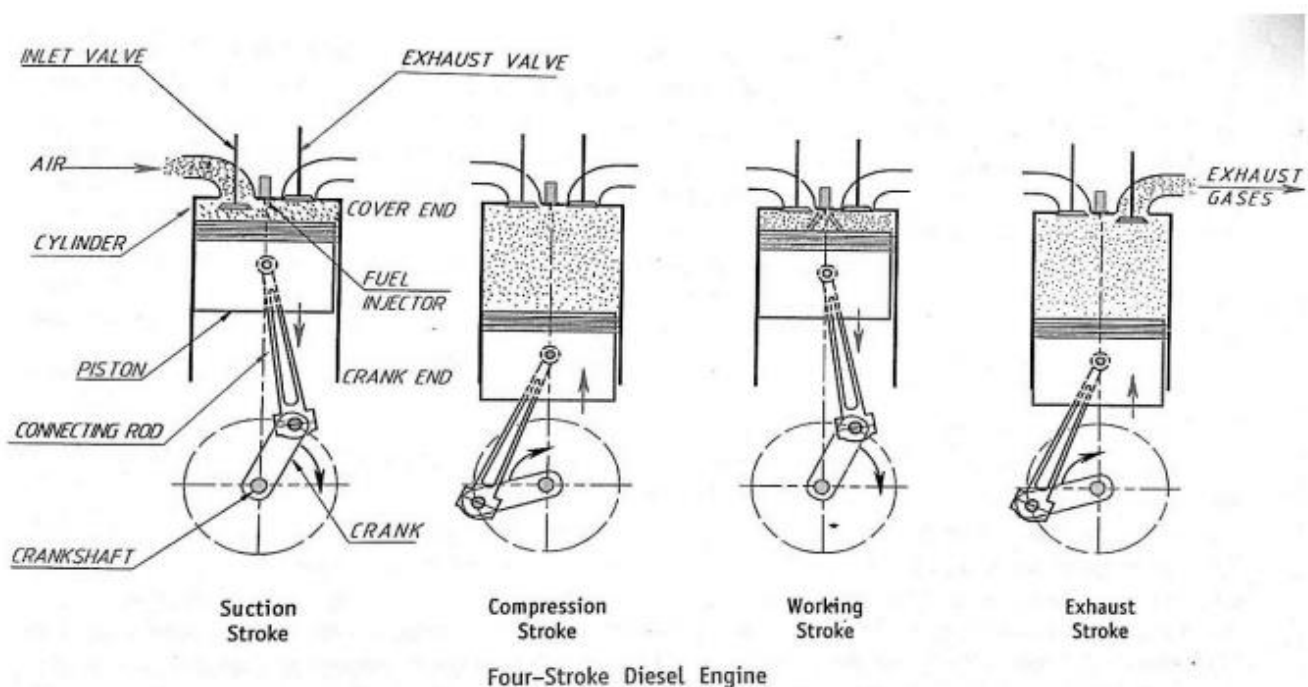
During this stroke the piston moves from BDC to TDC. The inlet valve is closed and exhaust valve is open. The crankshaft rotates by half a rotation. As the piston moves towards the TDC, the burnt gases will be expelled out through the exhaust valve. Line BA on the P-V diagram. When the piston reaches the TDC, the exhaust valve closes and this completes the cycle.

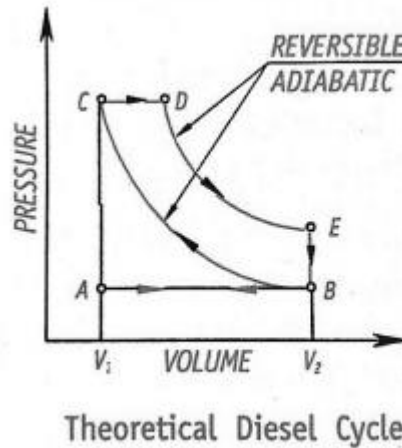
In 4 stroke engine, the 4 strokes constitute one cycle, hence the name 4 stroke cycle engine. The crankshaft makes two revolutions to complete one cycle. The power is developed in every alternate revolution of the crankshaft. 4 Stroke diesel engines produce higher power than 4 Stroke petrol engines.

4- STROKE DIESEL ENGINE (CI Engine):

Diesel engines works on the principle of theoretical Diesel cycle, also known as constant pressure cycle. It consists of cylinder, piston, connecting rod, crank, crankshaft, inlet valve, and exhaust valve and fuel injector. The fuel injector fitted at the top of the cylinder supplies the measured quantity of diesel at high pressure.

1. SUCTION STROKE: During this stroke the piston moves from TDC to BDC. The inlet valve is open and exhaust valve is closed. The crankshaft rotates by half a rotation. As the piston moves downwards fresh air is drawn into the cylinder through the inlet valve. At the end of this stroke, the piston is in BDC, the cylinder is filled with air and inlet valve closes. Horizontal line AB on the P-V diagram.





2. COMPRESSION STROKE: During this stroke the piston moves from BDC to TDC. Both the inlet valve and exhaust valves are closed. The crankshaft rotates by half a rotation. As the piston moves upwards, the air in the cylinder will be compressed. It is represented the BC curve in the P-V diagram. As the air gets compressed its pressure and temperature increases and attains a temperature greater than the ignition temperature of diesel. Diesel is sprayed into the cylinder through the fuel injector. The high temperature of the air ignites the diesel as soon as it is sprayed and undergoes combustion at constant pressure. Line CD on the P-V diagram. Since the compresses air ignites the diesel, this type of engine is also called as compression ignition or C.I Engine

3. WORKING OR POWER STROKE: During this stroke the piston moves from TDC to BDC. Both the inlet valve and exhaust valves are closed. The crankshaft rotates by half a rotation. The high pressure of the burnt gases forces the piston downwards performing power stroke. The linear motion of the piston is converted to rotary motion of the crankshaft by connecting rod and crank. It is represented by curve DE on PV diagram. At the end of the stroke, the piston is in BDC, the exhaust valve opens which release the burnt gases to the atmosphere. This will bring pressure in the cylinder to atmospheric at constant volume, represented by the line EB in the P-V diagram.

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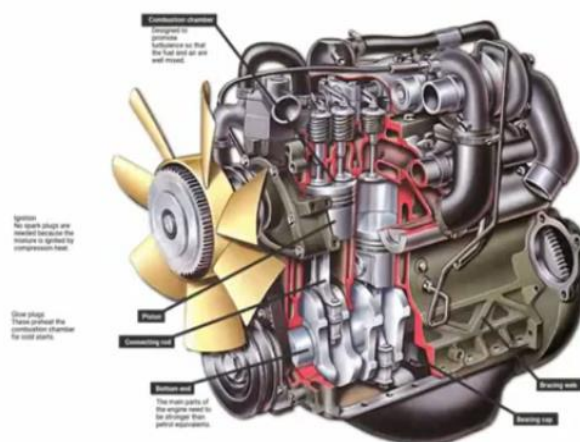
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COMPARISON OF PETROL AND DIESEL ENGINE

PRINCIPLE	PETROL	DIESEL
1. Fuel used	Petrol	Diesel
2. Admission of fuel	During suction stroke	At the end of compression
3. Charge drawn during	Air and petrol mixture	Only air
4. Type of ignition	Spark ignition	Compression
5. Uses	Scooter, motor cycle, car, etc.,	Trucks, tractors, buses, etc.
6. Engine speed	High about 7000rpm	Low from 500 to 3000rpm
7. Noise & vibration	Almost nil	High
8. Weight of the engine	Less	High
9. Initial cost	Less	More
10 Maintenance cost	Less	Slightly higher
11.Starting of the engine	Easily started	Difficult to start in cold
12. Exhaust gas pollution	More	More

Applications of IC engines

- Power generation
- Agriculture
- Marine
- Aircraft propulsion
- Automobile

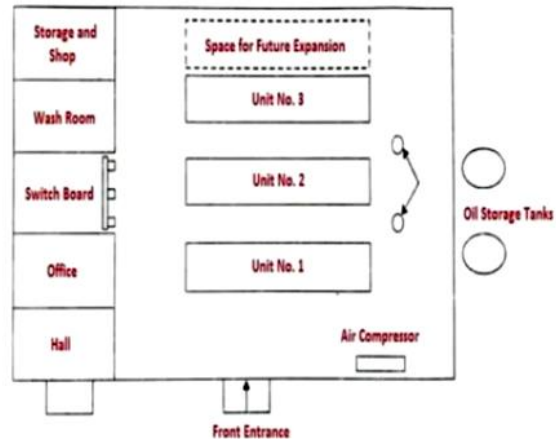


Application of IC engines **Power generation**

In a power plant, many diesel ICEs are grouped into blocks called generating unit sets. Every engine is connected to a shaft that is connected to its electric generator. These generating unit sets provide modular electric generating capacity and come in standardized sizes, ranging from 4 to 20 MW



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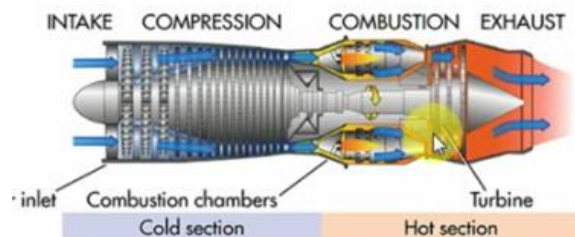


Layout of Diesel Power Plant

Aircraft Propulsion

The internal combustion (IC) engine is the power plant used on almost all light general aviation aircraft today. Modern aircrafts using the turbofan engines. Also electrical aircraft motors can promise a new and cleaner aviation future but are still a way off, powering prototypes. We will therefore focus on the internal combustion engine in this series discussing light aircraft propulsion.

JET ENGINE CROSS SECTION



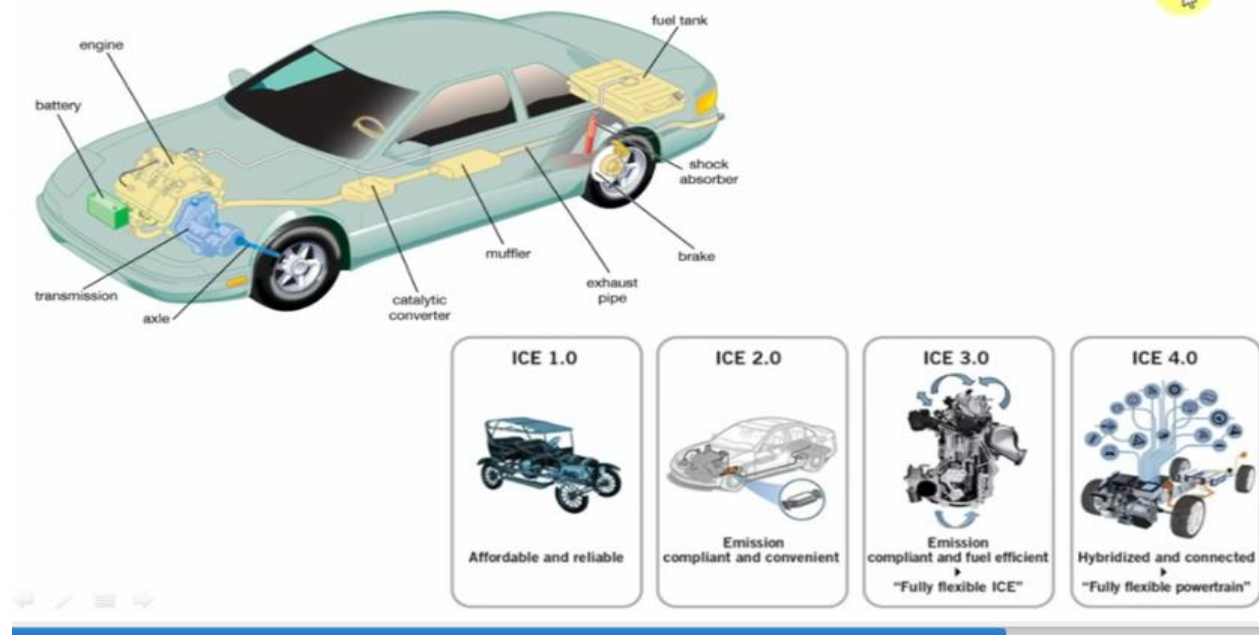
Jet type IC engine

Small aircrafts uses IC Engines



Automobile

Internal combustion (IC) engines are used in almost all automobiles or Road vehicles like scooters, motorcycles, cars, buses, trucks and heavy vehicles etc. The IC engine used may be petrol (SI) engine or Diesel(CI) engine.



Marine

Marine engines on ships are responsible for the propulsion of the vessel from one port to another. The engines used on board ships are internal combustion engines, in which, the combustion of fuel takes place inside the engine cylinder and the heat is generated post the combustion process. The ships of all types from goods to cruise and small boats are run by IC engines.



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Insight into Future Mobility; Electric and Hybrid Vehicles, Components of Electric and Hybrid Vehicles. Advantages and disadvantages of EVs and Hybrid vehicles.

Electric Vehicles

An electric vehicle (EV) is a vehicle that uses one or more electric motors for propulsion. It can be powered by a collector system, with electricity from extra vehicular sources, or it can be powered autonomously by a battery (sometimes charged by solar panels, or by converting fuel to electricity using fuel cells or a generator). EVs include, but are not limited to, road and rail vehicles, surface and underwater vessels, electric aircraft and electric spacecraft.

Below, you can see a summary of the key parts of an EV with simple definitions:

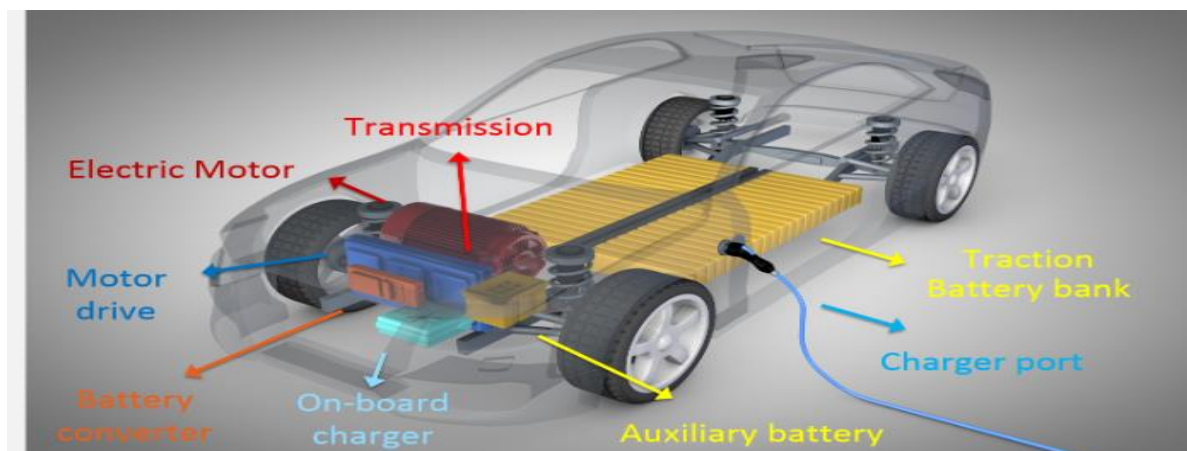
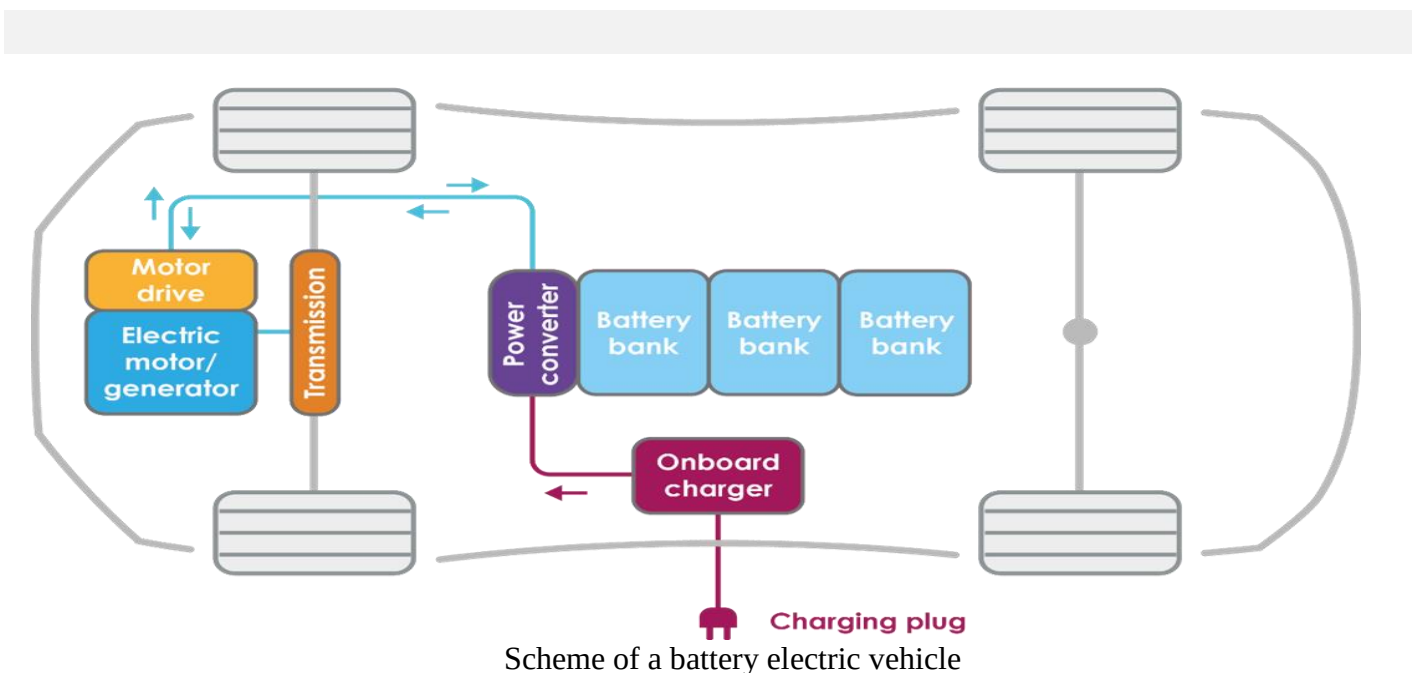


Figure: Key parts of a battery electric vehicle

Charging port or Vehicle inlet: It is a connector present on the electric vehicle to allow it to be connected to an external source of electricity for charging

On-board charger: It is an AC-to-DC power electronic converter (often referred to as a rectifier) that takes the incoming AC electricity supplied via the charge port and converts it to DC power for charging the traction battery. Using the battery management system, it regulates the battery characteristics such as voltage, current, temperature, and state of charge

Traction battery pack: It is a high voltage battery used to store energy in the electric car and provide power for use by the electric traction motor.

Battery power converter: It is a DC-to-DC power electronic converter that converts the voltage of the traction battery pack to the higher-voltage of the DC-bus used for power exchange with the traction motor.

Traction electric motor/generator: It is the main propulsion device in an electric car that converts electrical energy from the traction battery to mechanical energy for rotating the wheels. It also generates electricity by extracting energy from the rotating wheels while braking, and transferring that energy back to the traction battery pack.

Transmission: For an electric car, usually a single gear transmission with differential is used to transfer mechanical power from the traction motor to drive the wheels.

Advantages and Disadvantages of EVs and Hybrid vehicles

The electric car (EV) is a relatively new concept in the world of the automotive industry. Although some companies have based their entire model of cars around being proactive and using electricity, An electric car such as **TATA NEXON, Nissan Leaf, Ford Focus Electric or Tesla Model S, Chevrolet Volt** is a great way for you to not only save money but also help contribute towards a healthy and stable environment.

Advantages of an Electric Car

1. No Gas Required
2. More Convenient
3. Savings
4. No Emissions
5. Safe to Drive
6. Low Maintenance
7. Reduced Noise Pollution
8. Battery Life & Cost
9. Easy Driving
10. Regenerative Braking

Disadvantages of an Electric Car

1. Recharge Points
2. The Initial Investment is Steep
3. Short Driving Range and Speed
4. Longer Recharge Time
5. Silence as a Disadvantage
6. Battery Replacement
7. Not Suitable for Cities, Facing Shortage of Power
8. Minimal Amount of Pollution
9. Some Governments Do Not Provide Money Saving Initiatives to Encourage You to Buy an Electric Car

Advantages of an Electric Car

- 1. No Gas Required:** Electric cars are entirely charged by the electricity you provide, meaning you don't need to buy any gas ever again. Driving fuel-based cars can burn a hole in your pocket as prices of fuel have gone all-time high.
- 2. More Convenient:** The electric vehicle is easy to recharge, and the best part is you will no longer need to run to the fuel station to recharge your car before hitting the road! Even a normal household socket could be used for charging an electric car.
- 3. Savings:** These cars can be fuelled for very low prices, Electric cars can also be a great way to save money in your own life.
- 4. No Emissions:** The biggest advantage of an electric vehicle is its green credential. Electric cars are 100 percent eco-friendly as they run on electrically powered engines. It does not emit toxic gases or smoke in the environment as it runs on a clean energy source.
- 5. Safe to Drive:** Electric cars undergo the same fitness and testing procedures test as other fuel-powered cars. An electric car is safer to use, given their lower center of gravity, which makes them much more stable on the road in case of a collision. In case an accident occurs, one can expect airbags to open up and electricity supply to cut from the battery. This can prevent you and other passengers in the car from serious injuries.
- 6. Low Maintenance:** Electric cars run on electrically powered engines, and hence there is no need to lubricate the engines, anything related to the combustion engine or a ton of maintenance tasks that are usually associated with a gas engine. Other expensive engine work is a thing of the past. Therefore, the maintenance cost of these cars has come down. You don't need to send it to the service station often as you do for a standard gasoline-powered car.

7. Reduced Noise Pollution: Electric cars put a curb on noise pollution as they are much quieter. Electric motors are capable of providing smooth drive with higher acceleration over longer distances.

8. Battery Life & Cost: Batteries are an integral part of an electric vehicle. Most electric vehicle batteries are lithium ones, and their costs are improving every year.

The full capacity of a lithium-ion battery cell should be good for **300 to 500 cycles**. A good battery could last you up to **ten years**. With the improving technologies, the cost of these batteries is expected to come down even more.

9. Easy Driving: Electric cars have the simplest driving method. Electric cars come with a transmission comprising of only one really long gear. This effectively eliminates the need to add a clutch mechanism. Therefore, you can operate an electric car with just the accelerator pedal, brake pedal and steering wheel.

10. Regenerative Braking: In normal cars, the braking process is a total wastage of kinetic energy that gets released as frictional heat. However, in an electric vehicle, the same energy is used to charge the batteries.

Disadvantages of an Electric Car

Although the evidence of the positives has become very clear, there are also some downsides that each individual needs to consider before they decide to make an electric car their next big investment. These reasons are

1. Recharge Points: Electric fuelling stations are still in the development stages. Not a lot of places you go to on a daily basis will have electric fuelling stations for your vehicle, meaning that if you're on a long trip or decide to visit family in a rural or suburban area and run out of charge, it may be harder to find a charging station. You may be stuck where you are.

2. The Initial Investment is Steep: As EVs are very new, you may be surprised when you take a look at the price for EVs. Even the more affordable brands can be around Rs.20Lakhs to 30Lakhs.

3. Short Driving Range and Speed: Electric cars are limited by range and speed. Most of these cars have a range of about 50-100 miles (80-160km/hr) and need to be recharged again. You just can't use them for long journeys as of now, although it is expected to improve in the future.

4. Longer Recharge Time: While it takes a couple of minutes to fuel your gasoline-powered car, an electric car takes about 4-6 hours and sometimes even a day to get fully charged. Therefore, you need dedicated power stations as the time taken to recharge them is quite long.

5. Silence as a Disadvantage: Silence can be a bit disadvantage as people like to hear the noise if they are coming from behind them. An electric car is, however, silent and can lead to accidents in some cases.

6. Battery Replacement: Depending on the type and usage of battery, batteries of almost all electric cars are required to be changed every 3-10 years.

7. Not Suitable for Cities, Facing Shortage of Power: As electric cars need the power to charge up, the cities

that already facing acute power shortages are not suitable for electric cars. The consumption of more power would hamper their daily power needs.

8. Minimal Amount of Pollution: Electric vehicles are also not 100% emission-free; they cause a little amount of pollution indirectly. The batteries and electricity needed for charging are not necessarily generated from renewable energy sources

9. Some Governments Do Not Provide Money Saving Initiatives to Encourage You to Buy an Electric Car

Electric Vehicles in India: (Optional)

Best Range EVs Cost Per Kilometer			
Models	Range (Kms)	Cost Per Kilometer (₹)	
Hyundai Kona Electric	452	0.52	
Mercedes-Benz EQC	445	1.68	
MG ZS EV	340	0.62	
Tata Nexon EV	312	0.71	
Tata Tigor EV	213	1.42	
Mahindra e-Verito	181	1.67	
Mahindra e2o Plus	140	1.41	

EV parameters: Capacity, state of charge, range, energy consumption, power, torque (Optional)

1) Nominal battery capacity (E_{nom} , in Wh or kWh): It is **total electric energy that can be stored** in the battery. Alternately, it is the **maximum amount of electric energy that can be extracted** from a **fully charged battery state to the empty state**. Generally speaking, EV batteries have a battery capacity between **5 kWh to 100 kWh** depending on the type of EV. The higher the battery capacity, the more energy it can store and the **longer the time it takes to fully charge it**.

(Note: What does 60 kWh battery mean: Energy in Car Batteries To sum it up, a 60 kWh battery can generate 60 kW sustained over one hour . If you only use 15 kW per hour, you get four hours of energy. If you use more power, say 120 kW, your battery will only last for half an hour).

2. State of charge: (SOC , in %): The battery state of charge (SOC) is defined as the ratio between the amount of energy currently stored in the battery, (E_{batt}) to the total battery capacity, (E_{nom}) : $BSOC=(E_{batt}/E_{nom}) \times 100$.

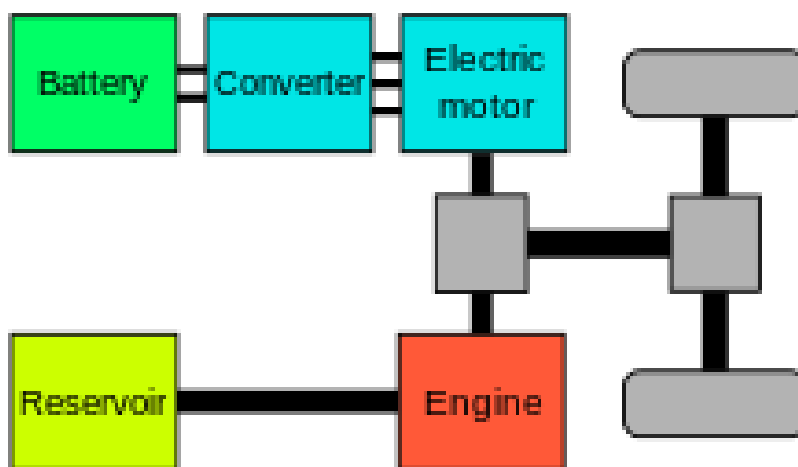
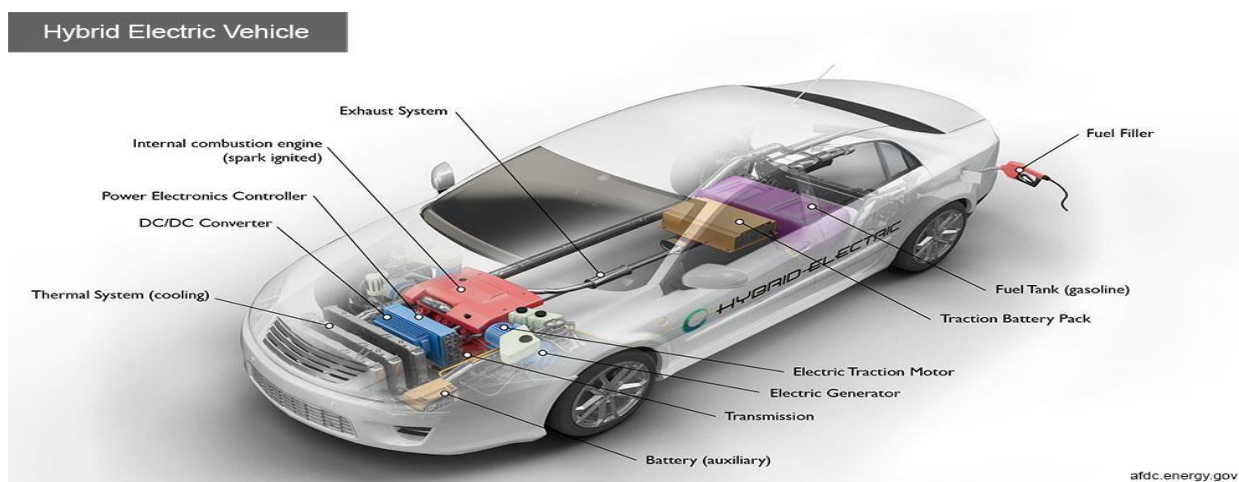
3. Range (R_{max} , in km): It is the **maximum distance** that can be driven by an electric car **when the battery is full**. Usually an electric car is tested using a standardized driving cycle to estimate

the range. The range can be expressed in **miles**, **kilometre** or other units based on the region. In this set of definitions, we stick to the European convention of using kilometre.

4. Motor power (P_m , in W): It is the **power delivered by the motor** to the wheels for propulsion. The motor power is positive or negative based on whether the car is driving or under regenerative braking. The motor power can be expressed as a product of the motor torque, T_m and the motor rotational speed, ω_m and the units normally used are watts (W), kilowatts (kW) or **horsepower (hp)**. **Note:** The equation is as simple as $hp = kW \times 1.369$, so a 100-kW motor produces 136 horsepower

Hybrid Electric vehicles

Hybrid electric vehicles are powered by an internal combustion engine and an electric motor, which uses energy stored in **batteries**. A hybrid electric vehicle cannot be plugged in to charge the battery. Instead, the battery is charged through regenerative braking and by the internal combustion engine. The extra power provided by the electric motor can potentially allow for a smaller engine. The battery can also power auxiliary loads and reduce engine idling when stopped. Together, these features result in better fuel economy without sacrificing performance



Block Diagram of Hybrid electric Car

Key Components of a Hybrid Electric Car

The internal combustion of the petrol engine continues to remain the primary source of power to the hybrid car. The electric motor derives power from regenerative braking; however, the hybrid's battery pack cannot recharge without the primary petrol engine.

Internal combustion engine (spark-ignited): In this configuration, fuel is injected into either the intake manifold or the combustion chamber, where it is combined with air, and the air/fuel mixture is ignited by the spark from a spark plug.

Traction Battery Pack: The pack stores electricity to power the electric motors and it also recharges through the electric generator.

Electric traction motor: This motor draws power from the battery pack and sends power to the wheels.

Electric generator: Generates electricity from the rotating wheels while braking, transferring that energy back to the traction battery pack

DC/DC converter: This device converts higher-voltage DC power from the traction battery pack to the lower-voltage DC power needed to run vehicle accessories and recharge the auxiliary battery.

Exhaust system: The exhaust system channels the exhaust gases from the engine out through the tailpipe.

Power electronics controller: This unit manages the flow of electrical energy delivered by the traction battery, controlling the speed of the electric traction motor and the torque it produces.

Thermal system (cooling): This system maintains a proper operating temperature range of the engine, electric motor, power electronics, and other components.

Transmission: The transmission transfers mechanical power from the engine and/or electric traction motor to drive the wheels.

Advantages and Disadvantages of Hybrid Cars:

Hybrid cars may be a stepping stone before the automobile industry moves into pure electric vehicles. So, understanding the pros and cons of hybrid cars will help you understand the technology.

Advantages/ Pros of Hybrid Cars:

Cleaner Emission: Compared to the internal combustion engine, hybrid cars employ both electric and internal combustion engines. The result is reduction in emissions and is environmentally friendly.

Less Fuel Dependency: With an electric motor to support the primary petrol engine, there is additional power available. Hence, there is less dependency on fossil fuel. **Smaller and Efficient Engine:** Since smaller engines don't have to power the hybrid car alone since there is an electric motor. Also, petrol engines used in hybrid cars are smaller in size and comparatively fuel efficient.

Regenerative Braking: Every time the brake is applied in a hybrid vehicle, the electric generator generates electricity and recharges the battery. This eliminates the need to stop the vehicle to charge the battery pack.

Disadvantages/Cons of Hybrid Cars:

Lower Performance: Since the main motive is to increase the fuel efficiency or range of the hybrid car, the power or acceleration can lag behind a conventional internal combustion engine car.

Expensive to Buy: Although car companies are trying to bridge the gap in pricing between a conventional vehicle and hybrid, hybrids continue to demand higher costs.

High Maintenance Cost With several mechanical parts in the cars and with two sets of engines powering the hybrids, the maintenance continues to be on the higher side. Also, not all mechanics are trained to repair a hybrid car.

Difference between Hybrid and Electric Cars:

Specifications	Hybrid Cars	Electric Cars
Power/Fuel Source	Electricity and Fossil Fuel (Petrol and Diesel)	Electricity Through Battery Pack (DC)
Engine	Internal Combustion Engine (ICE) and Electric Motor(s)	Electric Motor(s)
Fuel Efficiency	Combination of ICE and Battery Range	Depends on Battery Range
Emission Levels	Higher Compared to Electric Cars	Lower Compared to ICE and Hybrid Cars
Price Range	Similar to Conventional ICE Cars	High
Charging	Not Needed	Needed
Specifications	Hybrid Cars	Electric Cars

Hybrid Cars in India (Optional)

1. Maruti Suzuki 2. ErtigaMaruti Suzuki Ciaz 3. Maruti Suzuki Baleno	4. Toyota Glanz 5. Toyota Corolla 6. Hyundai Ioniq 7. Toyota Prius	8. Kia Niro 9. Hyundai Ioniq 10. Toyota C-HR
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